

Data Visualization

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1. Prepare the environment

```
load(file = "CHC9012_PCIBexDataPrepared.RData")

library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.5.2
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(Rmisc)

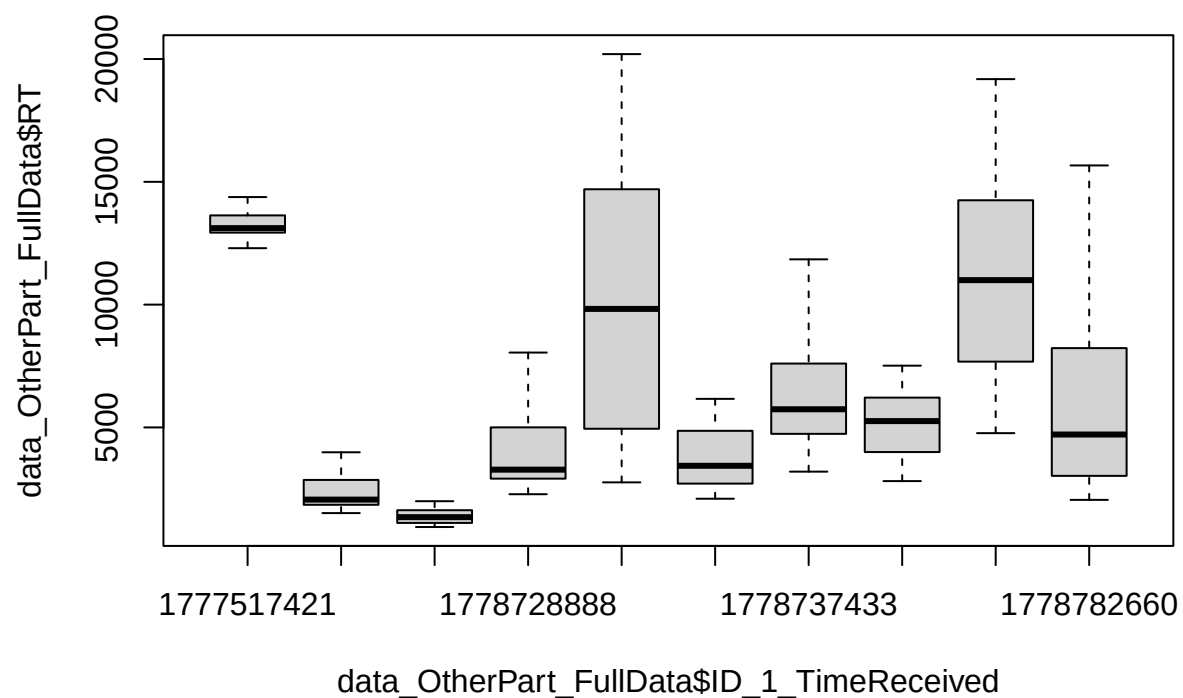
## Loading required package: lattice
## Loading required package: plyr
## -----
## You have loaded plyr after dplyr - this is likely to cause problems.
## If you need functions from both plyr and dplyr, please load plyr first, then dplyr:
## library(plyr); library(dplyr)
## -----
##
## Attaching package: 'plyr'
## The following objects are masked from 'package:dplyr':
##
##   arrange, count, desc, failwith, id, mutate, rename, summarise,
##   summarize
```

2. Cleaning the dataset

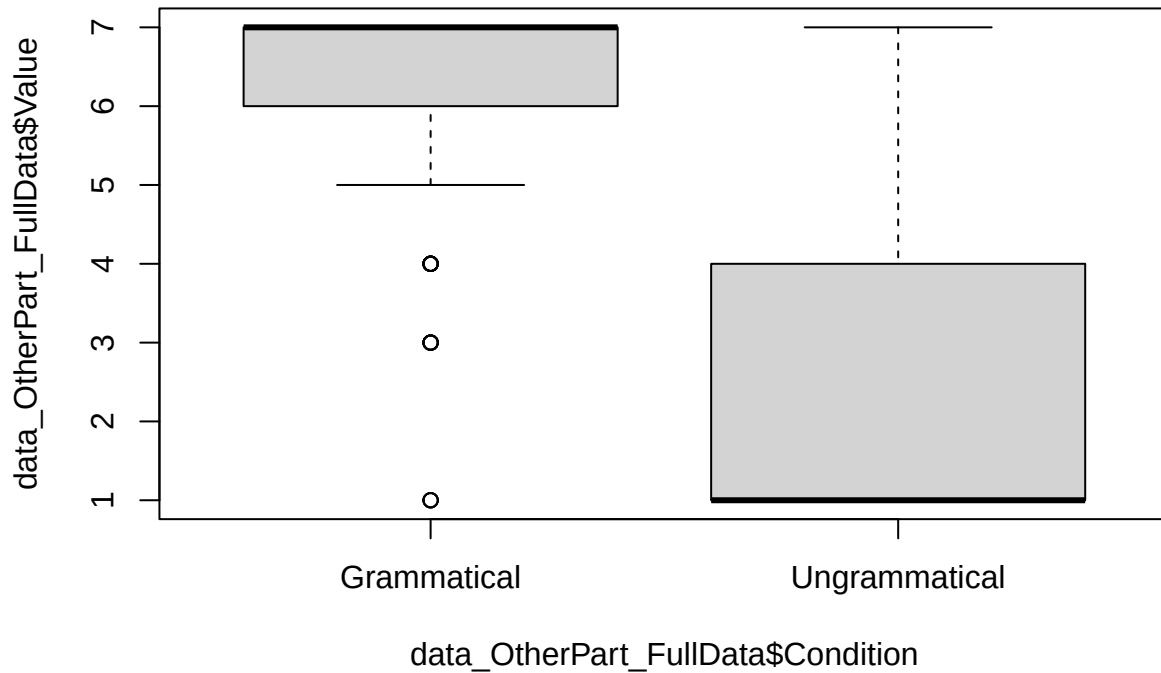
```
## 2.1 Select the participants with a human-like behavior
PartMeanRT <- aggregate(RT ~ ID_1_TimeReceived,
                        data = data3,
                        FUN = mean)

data_OtherPart <- subset(PartMeanRT, RT > 1000)
data_OtherPart_FullData <- droplevels(subset(data3,
                                             ID_1_TimeReceived %in% data_OtherPart$ID_1_TimeReceived))

plot(data_OtherPart_FullData$RT ~ data_OtherPart_FullData$ID_1_TimeReceived, outline = FALSE)
```



```
plot(data_OtherPart_FullData$Value ~ data_OtherPart_FullData$Condition)
```



```
## 2.2 Remove outlier trials
```

```
### 2.2.1 Absolute values
```

```
data_OtherPart_CleanAbsolute <- subset(data_OtherPart_FullData, RT > 200 & RT < 20000)
```

```
### 2.2.2 Relative values
```

```
# Calculate the mean per Participant and Condition
```

```
data_OtherPart_CleanAbsolute$m_rt <- ave(data_OtherPart_CleanAbsolute$RT,
                                         data_OtherPart_CleanAbsolute$ID_1_TimeReceived,
                                         data_OtherPart_CleanAbsolute$Condition,
                                         FUN = mean)
```

```
# 2. Calculate the SD per Participant and Condition
```

```
data_OtherPart_CleanAbsolute$sd_rt <- ave(data_OtherPart_CleanAbsolute$RT,
                                           data_OtherPart_CleanAbsolute$ID_1_TimeReceived,
                                           data_OtherPart_CleanAbsolute$Condition,
                                           FUN = sd)
```

```
# 3. Subset the data to remove outliers
```

```
data_OtherPart_CleanAbsoluteRelative <- subset(data_OtherPart_CleanAbsolute,
                                              RT >= (m_rt - 2.5 * sd_rt) &
                                              RT <= (m_rt + 2.5 * sd_rt))
```

```
### 2.2.3 Calculate the number of trials removed
```

```

FullData_NbTrials <- nrow(data_OtherPart_FullData)
TrimmedData_NbTrials <- nrow(data_OtherPart_CleanAbsoluteRelative)

# Number of trimmed trials
NbTrimmedTrials <- FullData_NbTrials - TrimmedData_NbTrials
NbTrimmedTrials

## [1] 14

# Proportion (%)
NbTrimmedTrials*100/FullData_NbTrials

## [1] 4.666667

```

3. Visualization of the clean results

3.1. Acceptability data

3.1.1 Fancy boxplots

```

# Create the canvas
ggplot(data_OtherPart_CleanAbsoluteRelative)

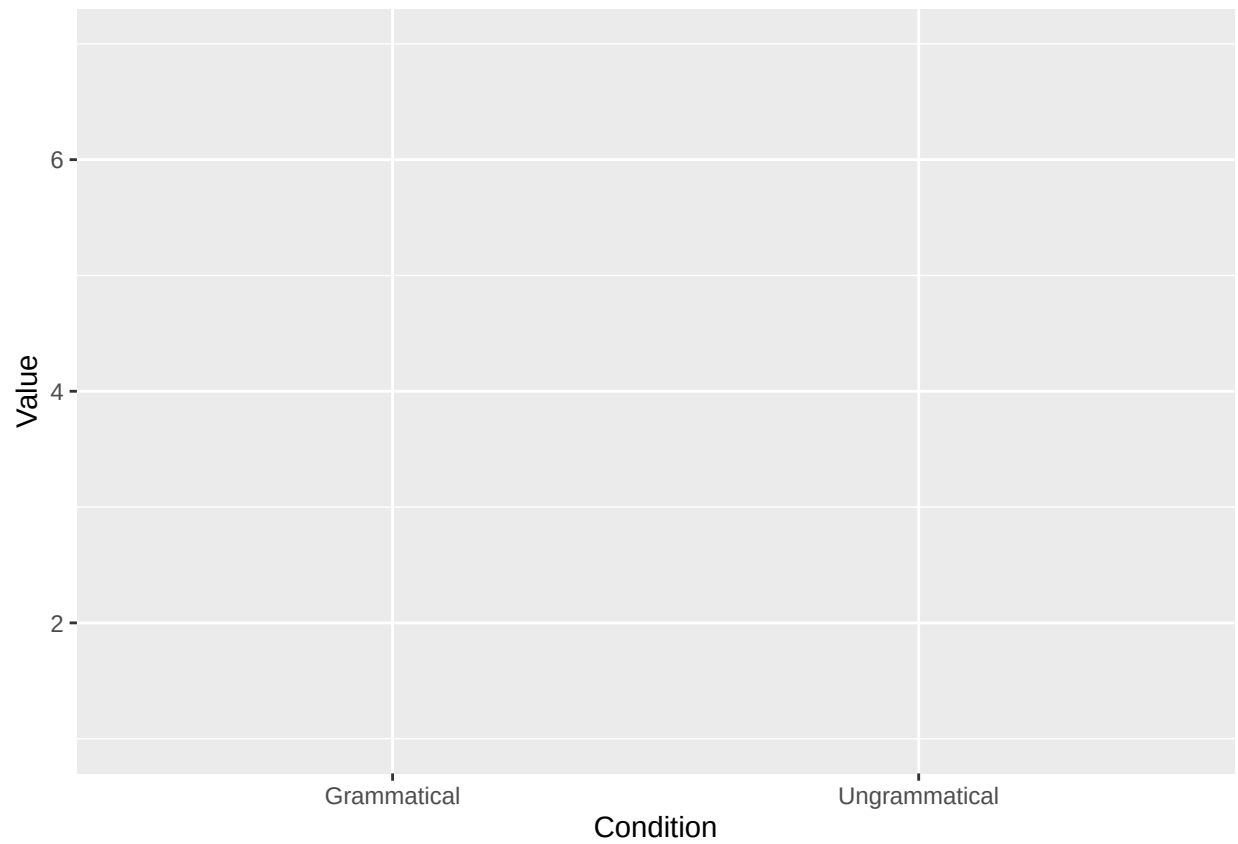
```

```

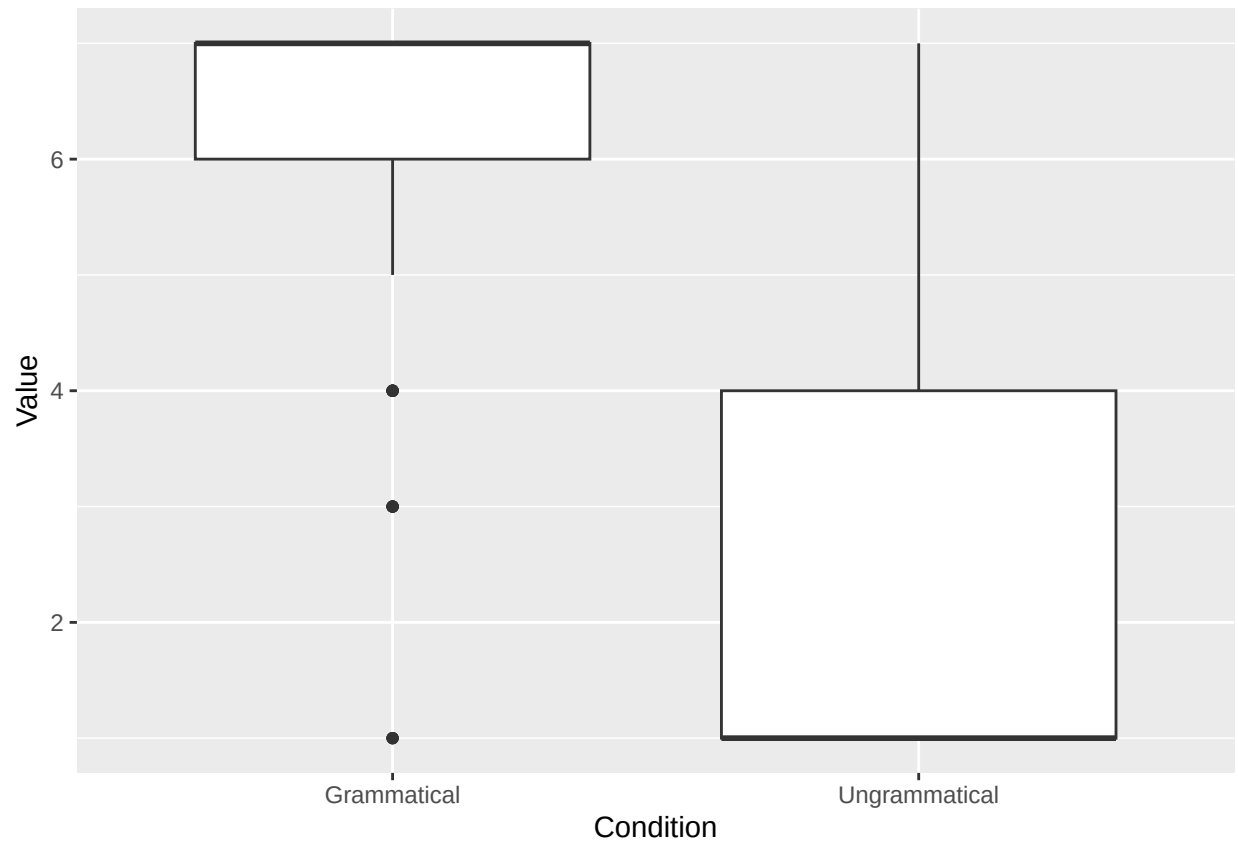
# Include what to plot (the "aesthetics")
ggplot(data_OtherPart_CleanAbsoluteRelative,
  aes(x = Condition,

```

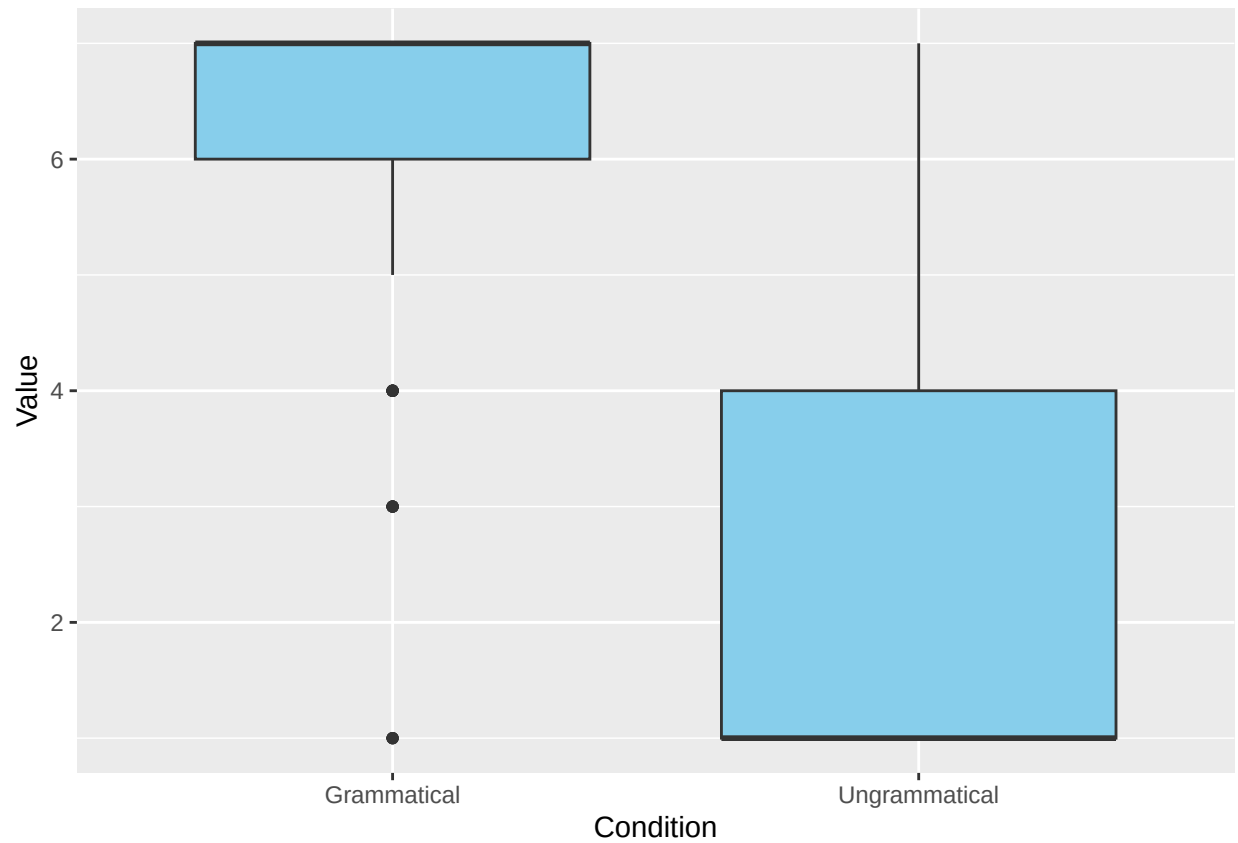
```
y = Value))
```



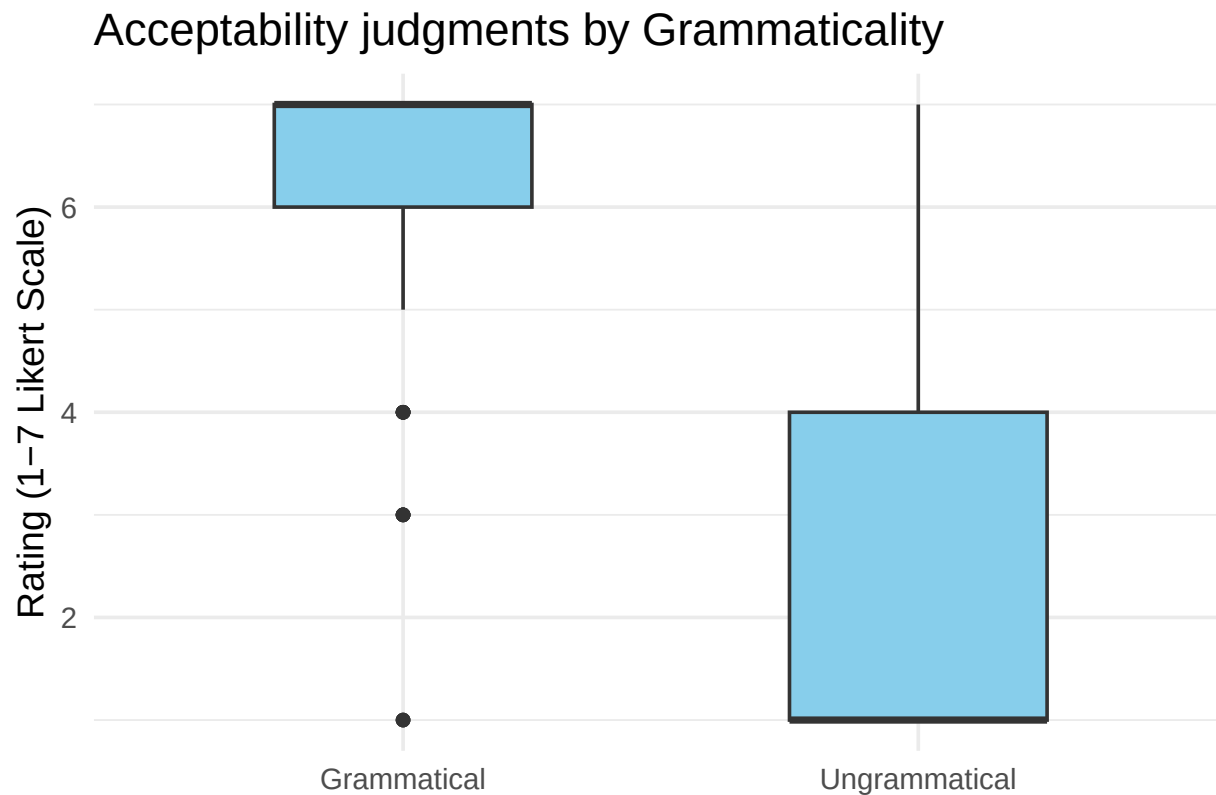
```
# Include how to plot  
ggplot(data_OtherPart_CleanAbsoluteRelative,  
  aes(x = Condition,  
    y = Value)) +  
  geom_boxplot()
```



```
# Add colors
ggplot(data_OtherPart_CleanAbsoluteRelative,
  aes(x = Condition,
    y = Value,
    fill = Condition)) +
  geom_boxplot(fill = "skyblue")
```

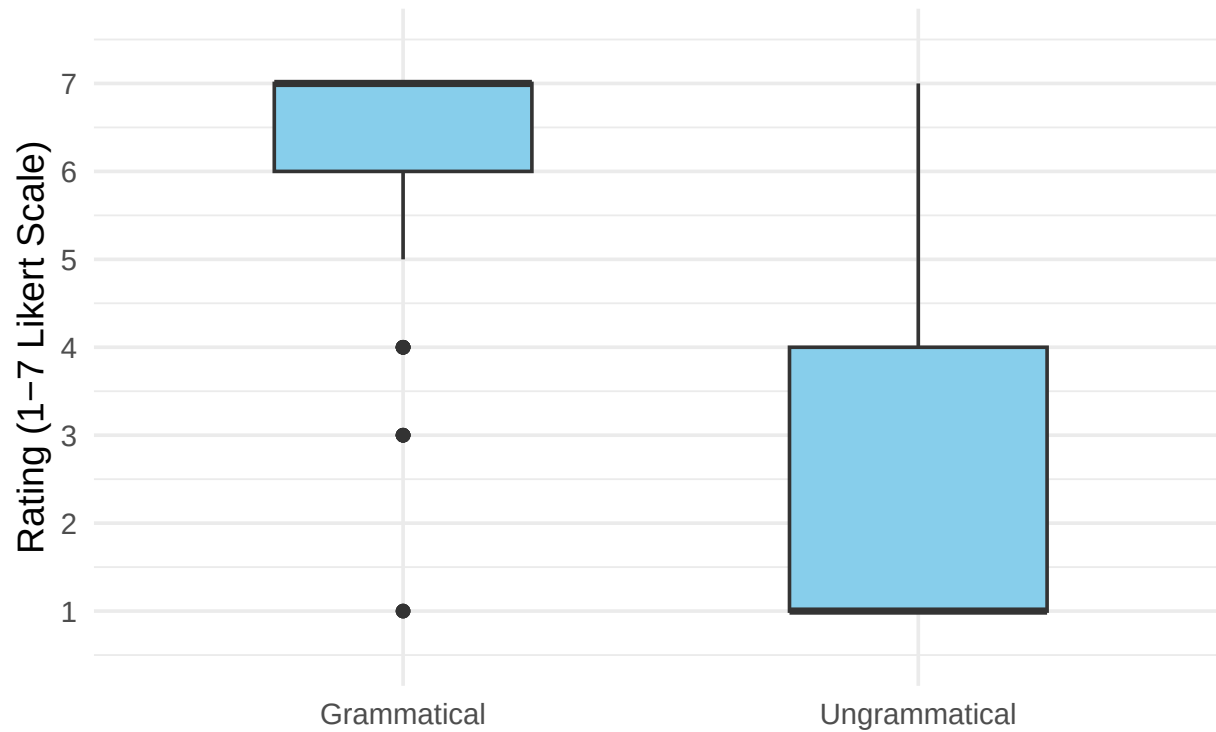


```
# Add title and legend, change the theme
ggplot(data_OtherPart_CleanAbsoluteRelative,
  aes(x = Condition,
    y = Value,
    fill = Condition)) +
  geom_boxplot(fill = "skyblue",
    width = 0.5) +
  labs(title = "Acceptability judgments by Grammaticality",
    x = "",
    y = "Rating (1-7 Likert Scale)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")
```



```
# Control the y axis
ggplot(data_OtherPart_CleanAbsoluteRelative,
  aes(x = Condition,
    y = Value,
    fill = Condition)) +
  geom_boxplot(fill = "skyblue",
    width = 0.5) +
  labs(title = "Acceptability judgments by Grammaticality",
    x = "",
    y = "Rating (1-7 Likert Scale)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none") +
  scale_y_continuous(limits = c(0.5, 7.5),
    breaks = 1:7)
```

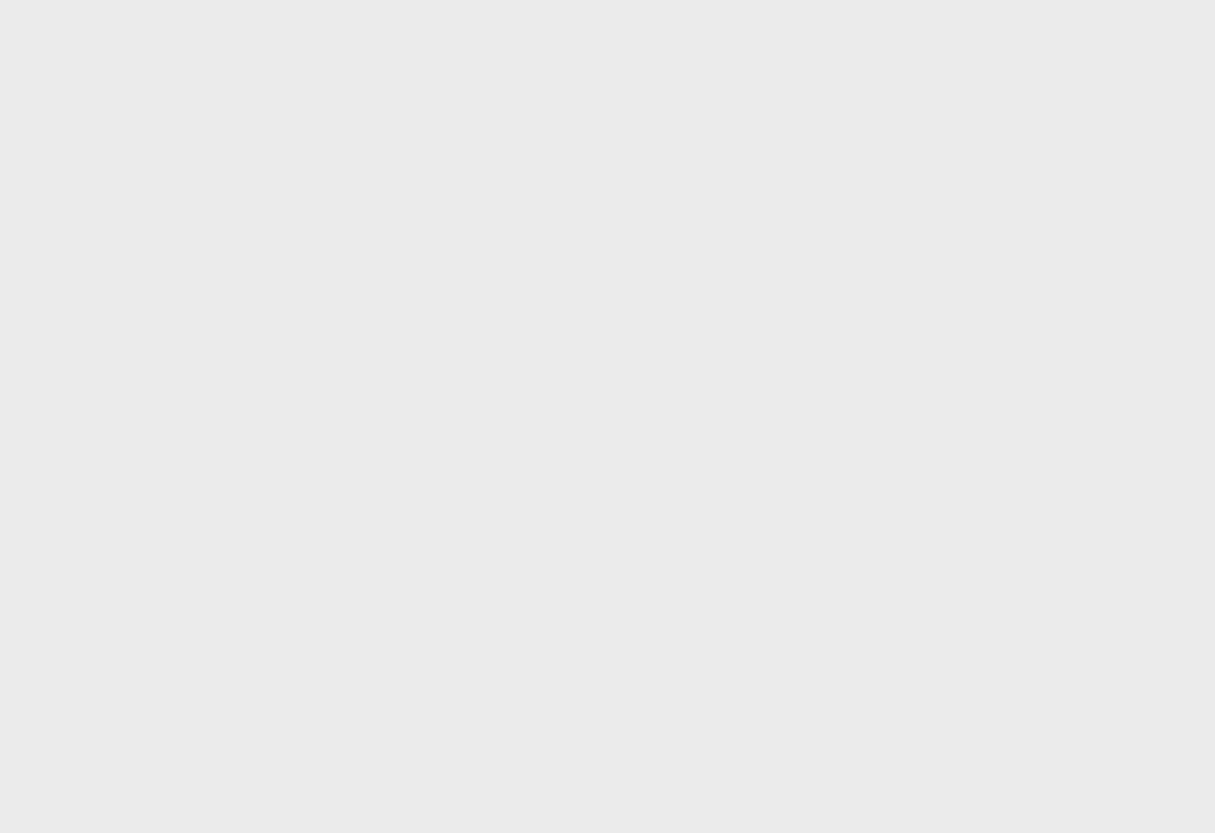
Acceptability judgments by Grammaticality



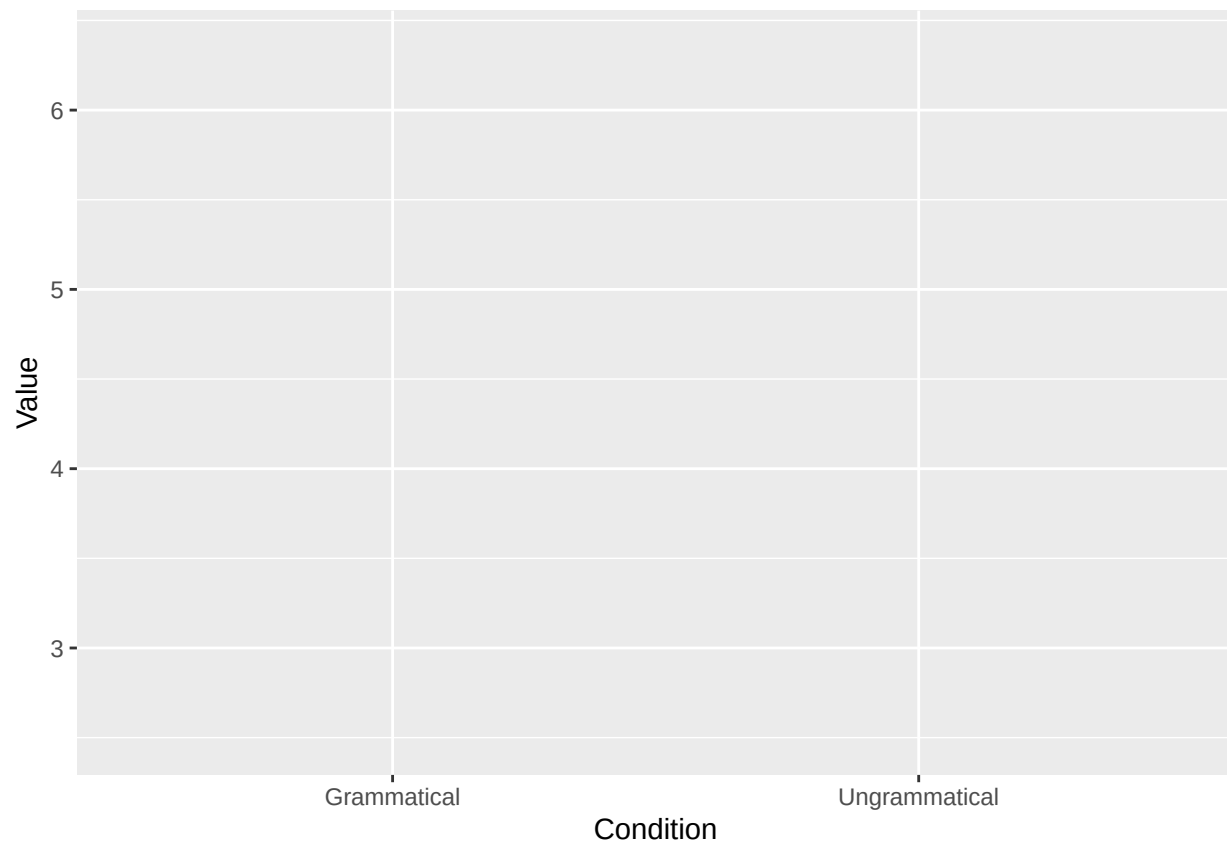
3.1.2 Bar plots

```
JudgSummary <- summarySE(data_OtherPart_CleanAbsoluteRelative,  
                           measurevar = "Value",  
                           groupvars = "Condition")
```

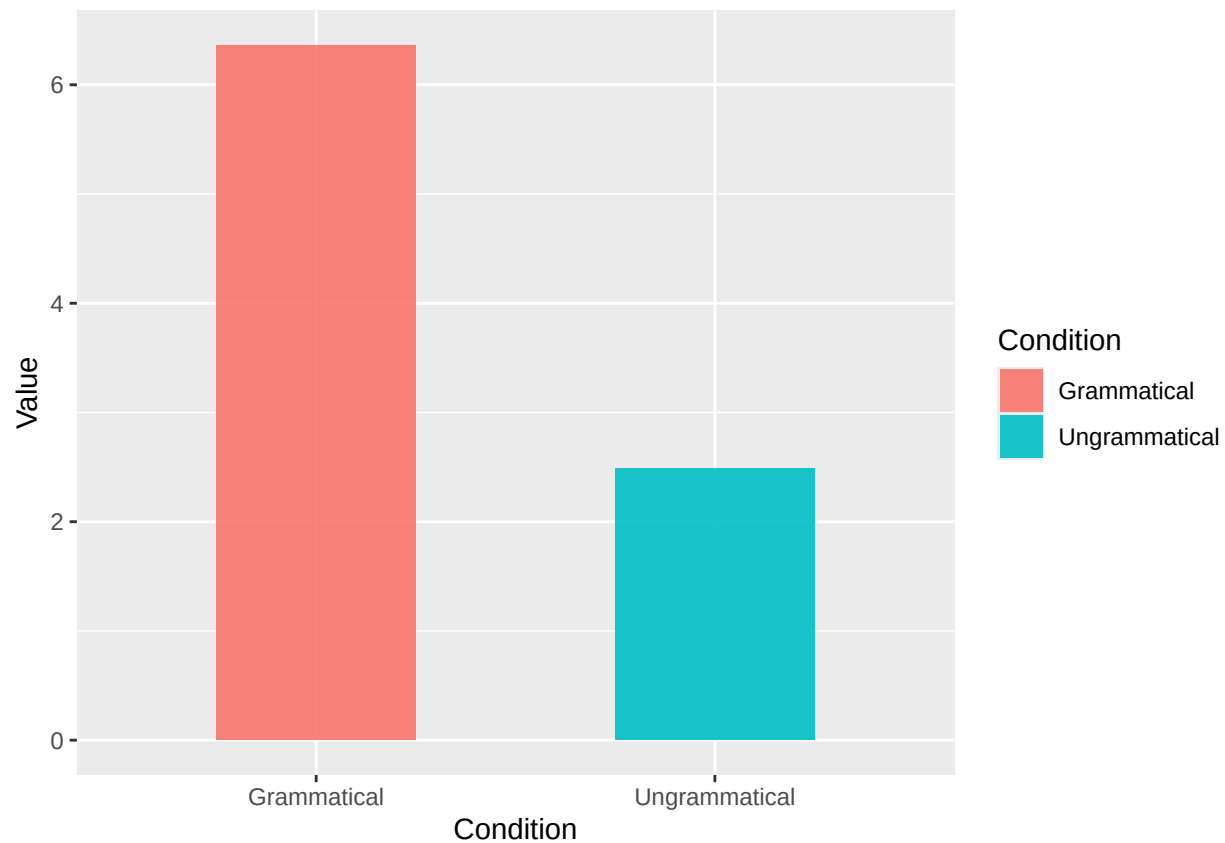
```
# Create the canvas  
ggplot(JudgSummary)
```



```
# Include what to plot (the "aesthetics")
ggplot(JudgSummary,
  aes(x = Condition,
    y = Value,
    fill = Condition))
```

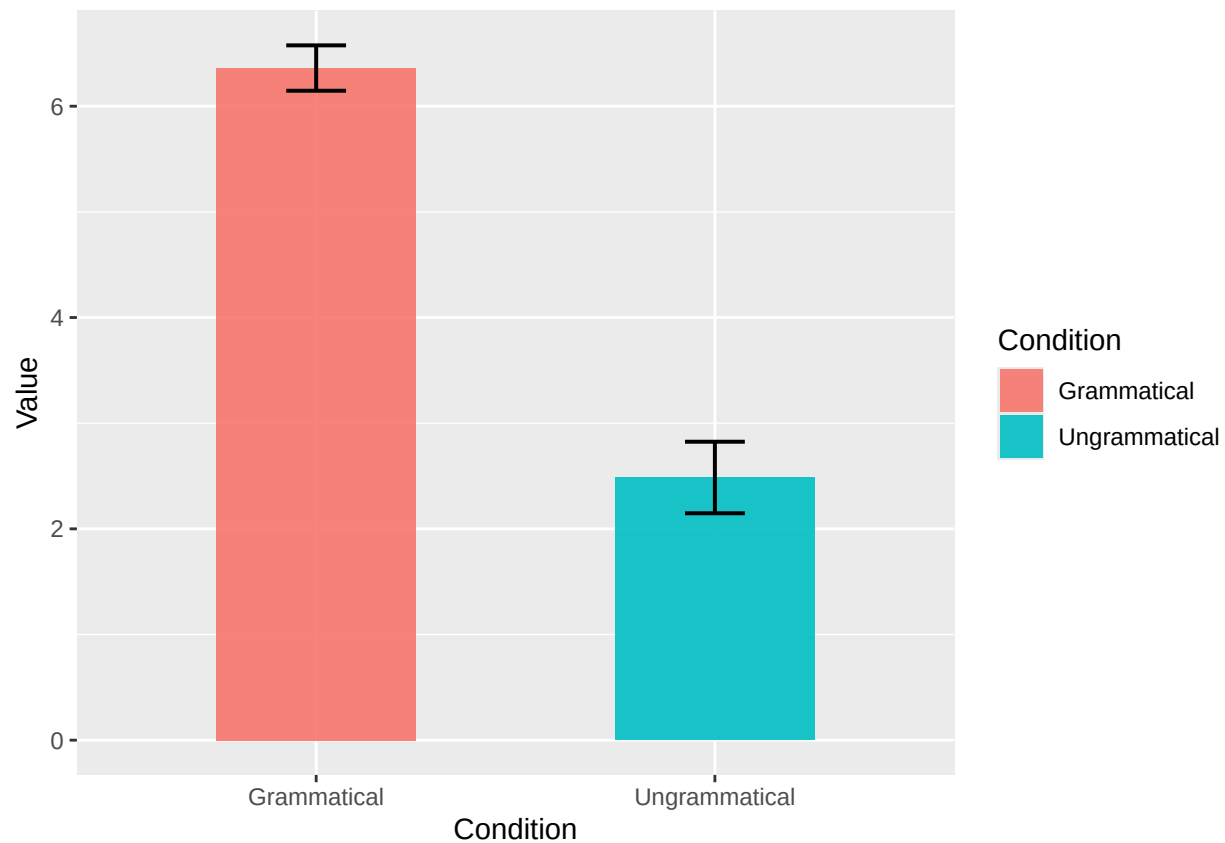


```
# Include how to plot  
ggplot(JudgSummary,  
  aes(x = Condition,  
    y = Value,  
    fill = Condition)) +  
  geom_col(width = 0.5, alpha = 0.9)
```

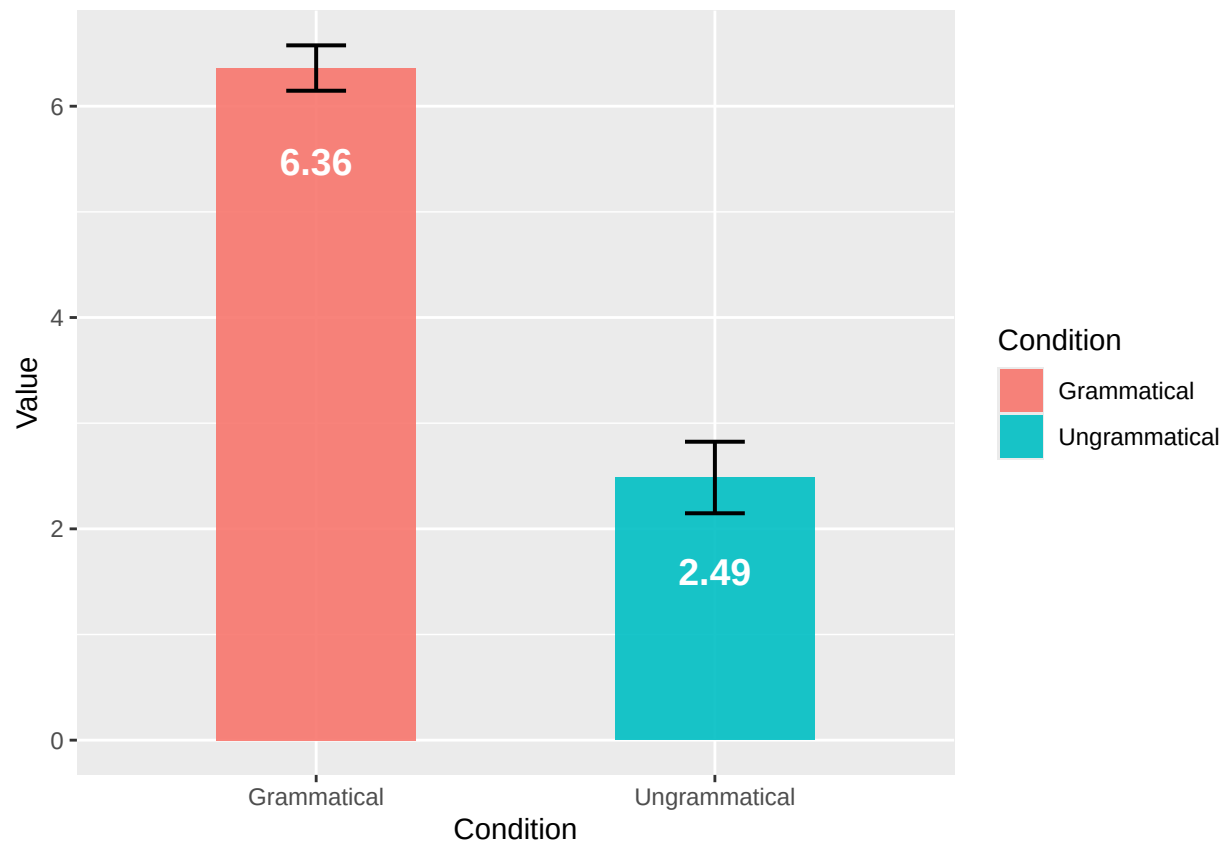


```
# Include error bars
ggplot(JudgSummary,
  aes(x = Condition,
    y = Value,
    fill = Condition)) +
  geom_col(width = 0.5, alpha = 0.9) +
  geom_errorbar(aes(ymin = Value - ci,
    ymax = Value + ci),
    width = 0.15,
    size = 0.7)
```

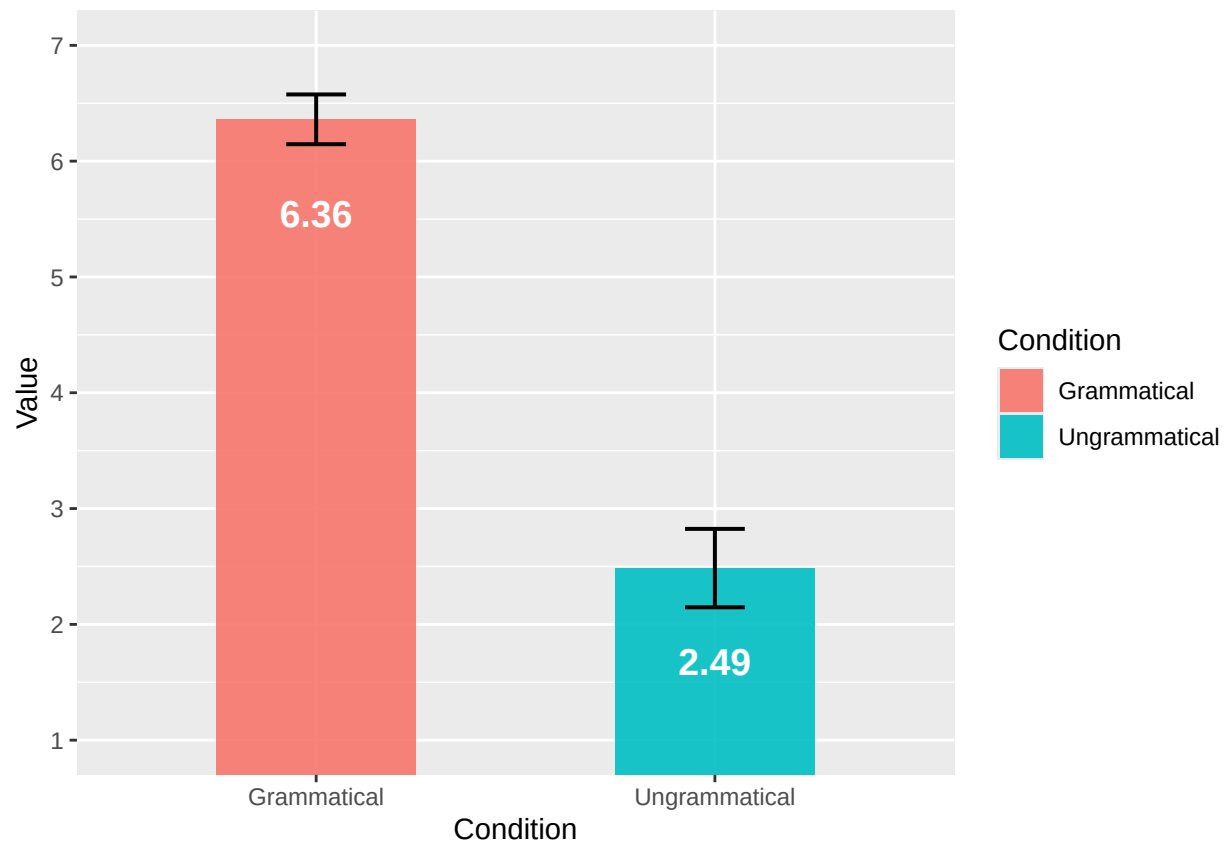
```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



```
# Include text
ggplot(JudgSummary,
  aes(x = Condition,
    y = Value,
    fill = Condition)) +
  geom_col(width = 0.5, alpha = 0.9) +
  geom_errorbar(aes(ymin = Value - ci,
    ymax = Value + ci),
    width = 0.15,
    size = 0.7) +
  geom_text(aes(label = round(Value, 2)),
    vjust = 4,
    color = "white",
    size = 5,
    fontface = "bold")
```



```
# Control the y axis
ggplot(JudgSummary,
  aes(x = Condition,
      y = Value,
      fill = Condition)) +
  geom_col(width = 0.5, alpha = 0.9) +
  geom_errorbar(aes(ymin = Value - ci,
                    ymax = Value + ci),
    width = 0.15,
    size = 0.7) +
  geom_text(aes(label = round(Value, 2)),
    vjust = 4,
    color = "white",
    size = 5,
    fontface = "bold") +
  scale_y_continuous(limits = c(1, 7),
    breaks = 1:7,
    oob = scales::rescale_none)
```

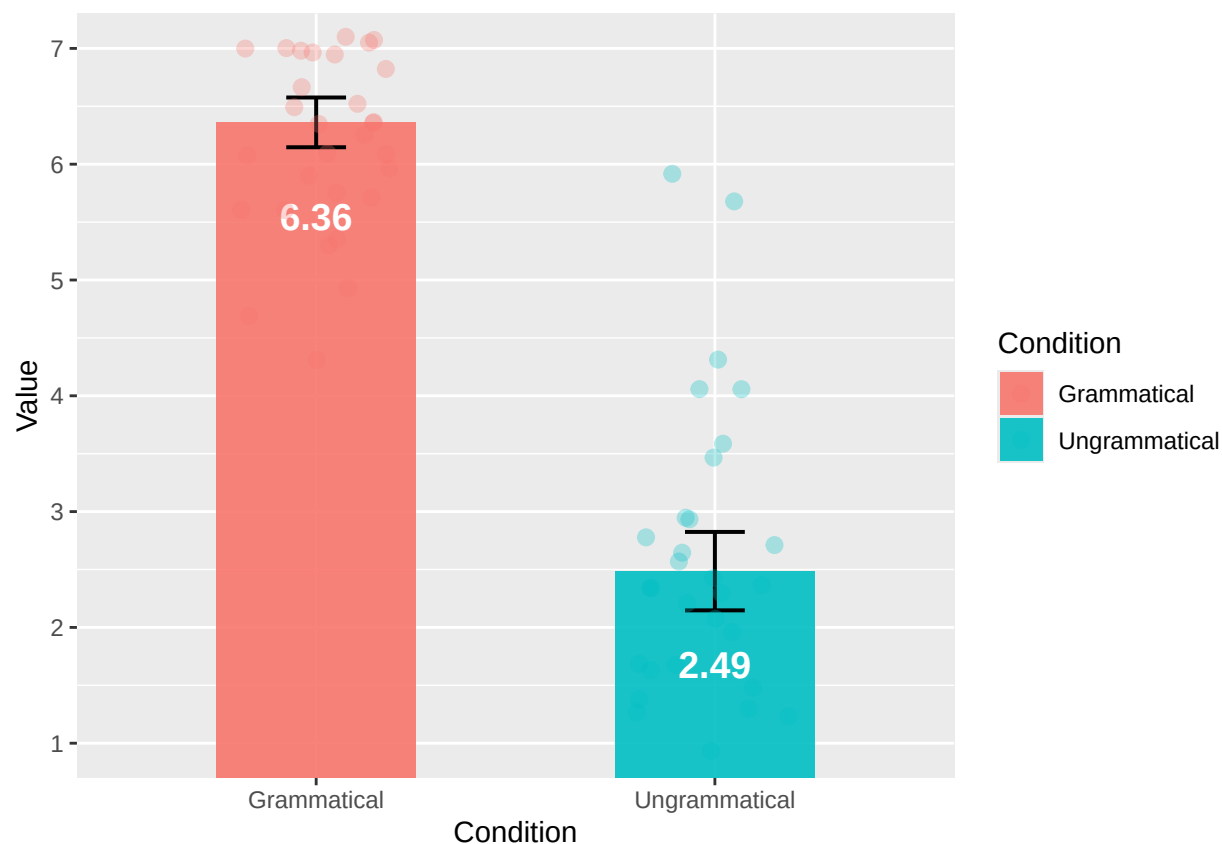


```
# Add information about individual items

JudgSummaryItem <- summarySE(data_OtherPart_CleanAbsoluteRelative,
                             measurevar = "Value",
                             groupvars = c("Condition", "ItemNo"))

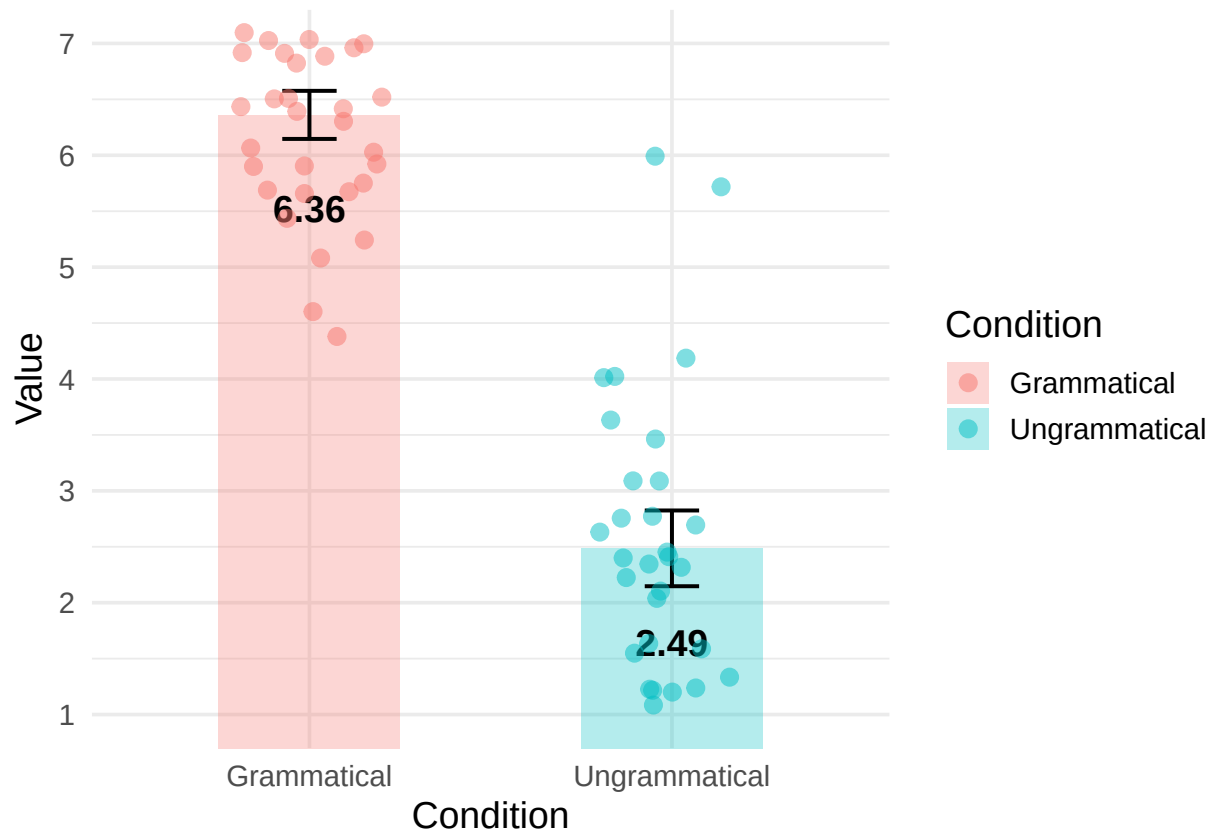
ggplot(JudgSummary,
       aes(x = Condition,
           y = Value,
           fill = Condition)) +
  geom_col(width = 0.5, alpha = 0.9) +
  geom_errorbar(aes(ymin = Value - ci,
                   ymax = Value + ci),
               width = 0.15,
               size = 0.7) +
  geom_text(aes(label = round(Value, 2)),
            vjust = 4,
            color = "white",
            size = 5,
            fontface = "bold") +
  scale_y_continuous(limits = c(1, 7),
                    breaks = 1:7,
                    oob = scales::rescale_none) +
  geom_jitter(data = JudgSummaryItem,
             aes(color = Condition),
             width = 0.2,
```

```
height = 0.1,
alpha = 0.3,
size = 2.5)
```



```
# Change the *shade* of the columns, the *color* of the text, and the theme to "minimal"
ggplot(JudgSummary,
  aes(x = Condition,
      y = Value,
      fill = Condition)) +
  geom_col(width = 0.5, alpha = 0.3) +
  geom_errorbar(aes(ymin = Value - ci,
                    ymax = Value + ci),
                width = 0.15,
                size = 0.7) +
  geom_text(aes(label = round(Value, 2)),
            vjust = 4,
            color = "black",
            size = 5,
            fontface = "bold") +
  scale_y_continuous(limits = c(1, 7),
                    breaks = 1:7,
                    oob = scales::rescale_none) +
  geom_jitter(data = JudgSummaryItem,
              aes(color = Condition),
              width = 0.2,
              height = 0.1,
```

```
alpha = 0.5,
size = 2.5) +
theme_minimal(base_size = 14)
```

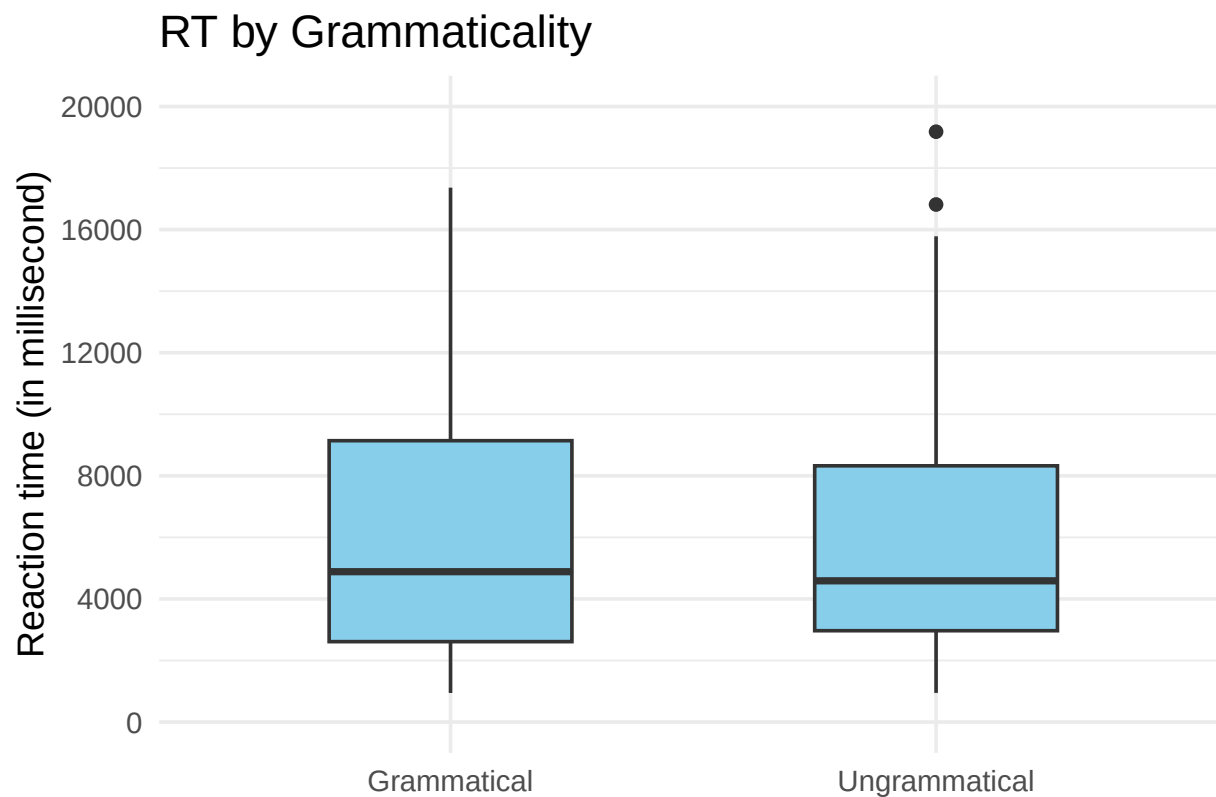


3.1 Visualizing the background information

3.2 Reaction time

3.2.1 Fancy boxplots

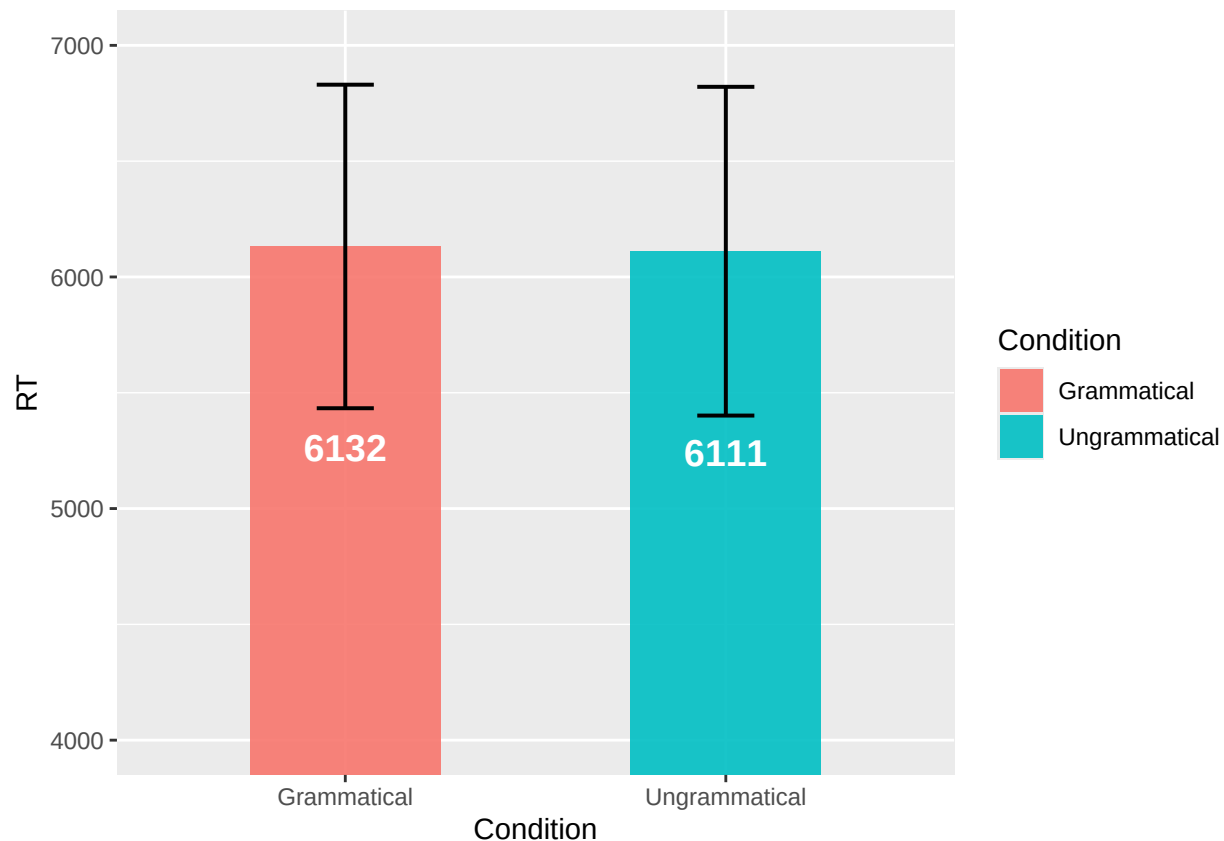
```
ggplot(data_OtherPart_CleanAbsoluteRelative,
       aes(x = Condition,
           y = RT,
           fill = Condition)) +
  geom_boxplot(fill = "skyblue",
              width = 0.5) +
  labs(title = "RT by Grammaticality",
       x = "",
       y = "Reaction time (in millisecond)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none") +
  scale_y_continuous(limits = c(0, 20000),
                    breaks = seq(0, 20000, by = 4000))
```



3.2.2 Bar plots

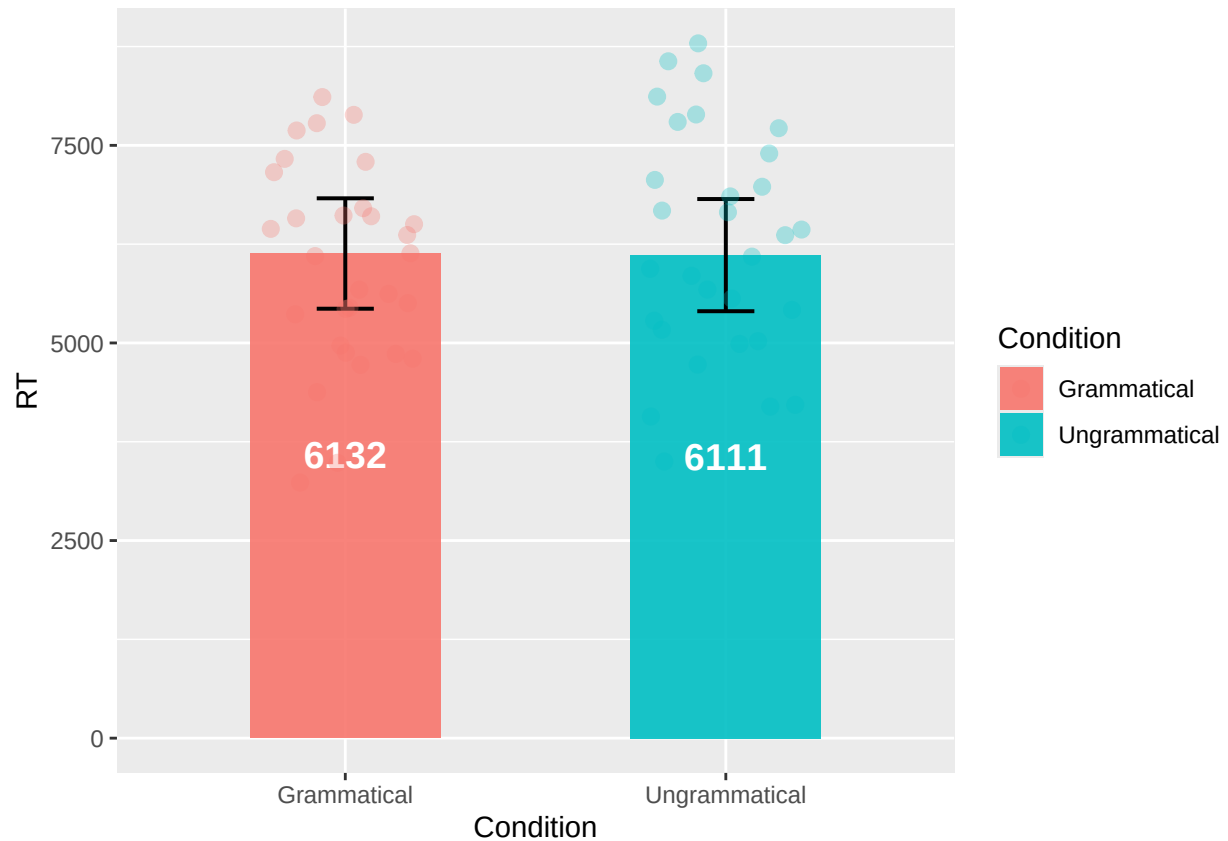
```
RTSummary <- summarySE(data_OtherPart_CleanAbsoluteRelative,
  measurevar = "RT",
  groupvars = "Condition")
```

```
ggplot(RTSummary,
  aes(x = Condition,
    y = RT,
    fill = Condition)) +
  geom_col(width = 0.5, alpha = 0.9) +
  geom_errorbar(aes(ymin = RT - ci,
    ymax = RT + ci),
    width = 0.15,
    size = 0.7) +
  geom_text(aes(label = round(RT)),
    vjust = 8,
    color = "white",
    size = 5,
    fontface = "bold") +
  scale_y_continuous(limits = c(4000, 7000),
    breaks = seq(4000, 7000, by = 1000),
    oob = scales::rescale_none)
```

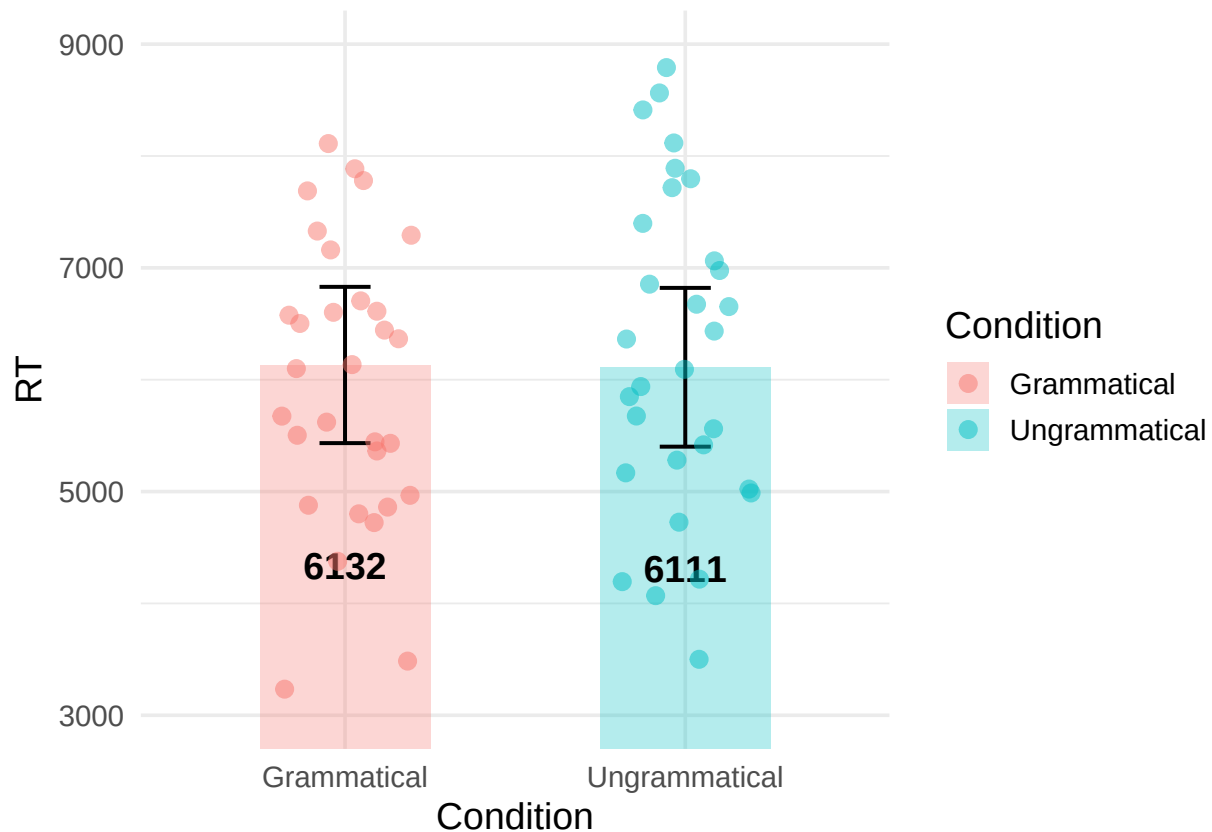


```
# Add information about individual items
RTSummaryItem <- summarySE(data_OtherPart_CleanAbsoluteRelative,
  measurevar = "RT",
  groupvars = c("Condition", "ItemNo"))

# Plot the figure
ggplot(RTSummary,
  aes(x = Condition,
    y = RT,
    fill = Condition)) +
  geom_col(width = 0.5, alpha = 0.9) +
  geom_errorbar(aes(ymin = RT - ci,
    ymax = RT + ci),
    width = 0.15,
    size = 0.7) +
  geom_text(aes(label = round(RT)),
    vjust = 8,
    color = "white",
    size = 5,
    fontface = "bold") +
  geom_jitter(data = RTSummaryItem,
    aes(color = Condition),
    width = 0.2,
    height = 0.1,
    alpha = 0.3,
    size = 2.5)
```



```
# Change the *shade* of the columns, the *color* of the text, and the theme to "minimal"
ggplot(RTSummary,
  aes(x = Condition,
      y = RT,
      fill = Condition)) +
  geom_col(width = 0.5, alpha = 0.3) +
  geom_errorbar(aes(ymin = RT - ci,
                    ymax = RT + ci),
    width = 0.15,
    size = 0.7) +
  geom_text(aes(label = round(RT)),
    vjust = 8,
    color = "black",
    size = 5,
    fontface = "bold") +
  scale_y_continuous(limits = c(3000, 9000),
    breaks = seq(3000, 9000, by = 2000),
    oob = scales::rescale_none) +
  geom_jitter(data = RTSummaryItem,
    aes(color = Condition),
    width = 0.2,
    height = 0.1,
    alpha = 0.5,
    size = 2.5) +
  theme_minimal(base_size = 14)
```

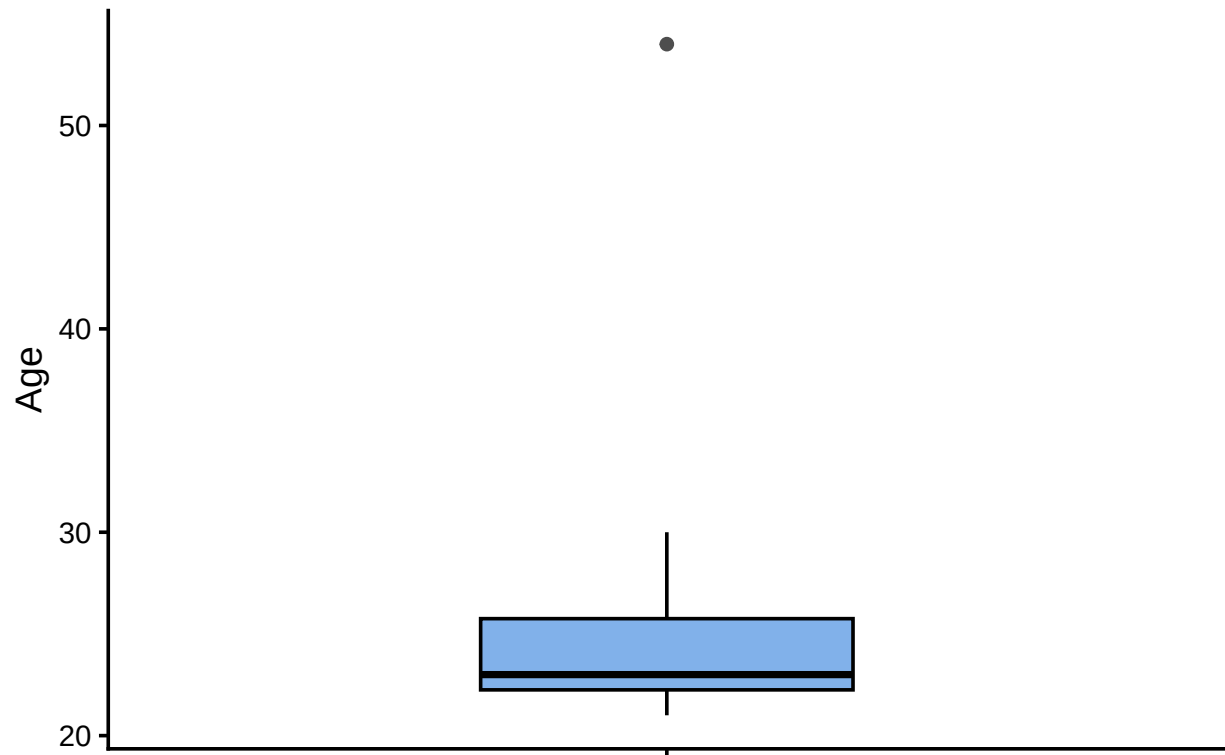


3.3 Look at the background information

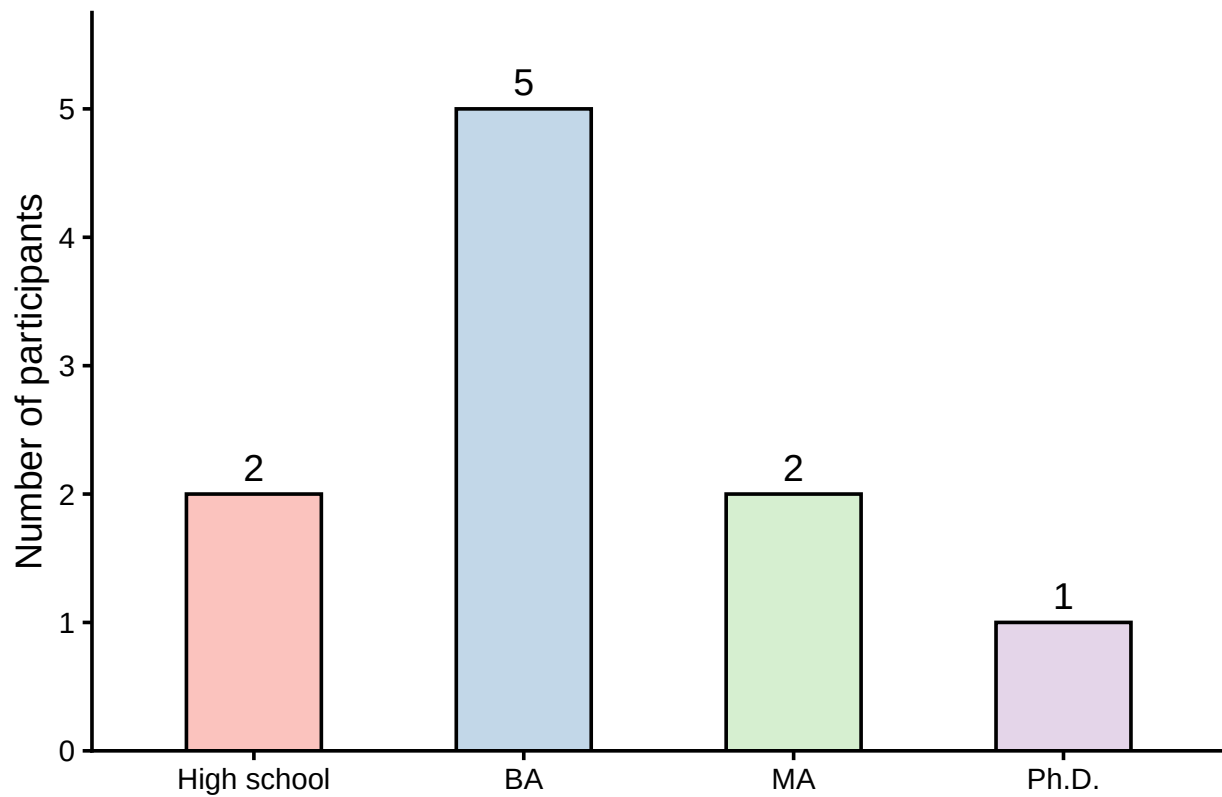
```
# Summarize the background information by participant
ParticipantBackground <- data_OtherPart_CleanAbsoluteRelative %>%
  distinct(ID_1_TimeReceived, .keep_all = TRUE) %>%
  select(ID_1_TimeReceived, UserAge, UserDegree)

ParticipantBackground$UserDegree <- factor(ParticipantBackground$UserDegree,
  levels = c("High school", "BA", "MA", "Ph.D.") )

# Plot the age of the participants
ggplot(ParticipantBackground, aes(x = "", y = UserAge)) +
  geom_boxplot(
    width = 0.4,
    fill = "#4a90e2",      # Soft academic blue
    color = "black",       # Crisp border line
    alpha = 0.7
  ) +
  theme_classic(base_size = 14) +
  labs(x = "",
    y = "Age") +
  theme(legend.position = "none")
```



```
# Plot the degree of the participants
ggplot(ParticipantBackground,
  aes(x = UserDegree,
    fill = UserDegree)) +
  geom_bar(width = 0.5,
    color = "black",
    alpha = 0.8,
    show.legend = FALSE) +
  geom_text(stat = "count",
    aes(label = after_stat(count)),
    vjust = -0.5,
    size = 5) +
  labs(x = "",
    y = "Number of participants") +
  scale_fill_brewer(palette = "Pastel1") +
  theme_classic(base_size = 14) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15)))
```



Plot two types of information at the same time

```
ggplot(ParticipantBackground,  
  aes(x = UserDegree,  
      y = UserAge,  
      fill = UserDegree)) +  
  geom_jitter(width = 0.1,  
             height = 0,  
             alpha = 0.5,  
             size = 2.5,  
             color = "dimgray") +  
  geom_boxplot(width = 0.4,  
              alpha = 0.7,  
              outlier.shape = NA) +  
  labs(x = "Highest degree",  
       y = "Age") +  
  scale_fill_brewer(palette = "Pastell1") +  
  theme_classic(base_size = 14) +  
  theme(legend.position = "none")
```

